

Superposition Theorem

Dr. Sathya P
Associate Professor
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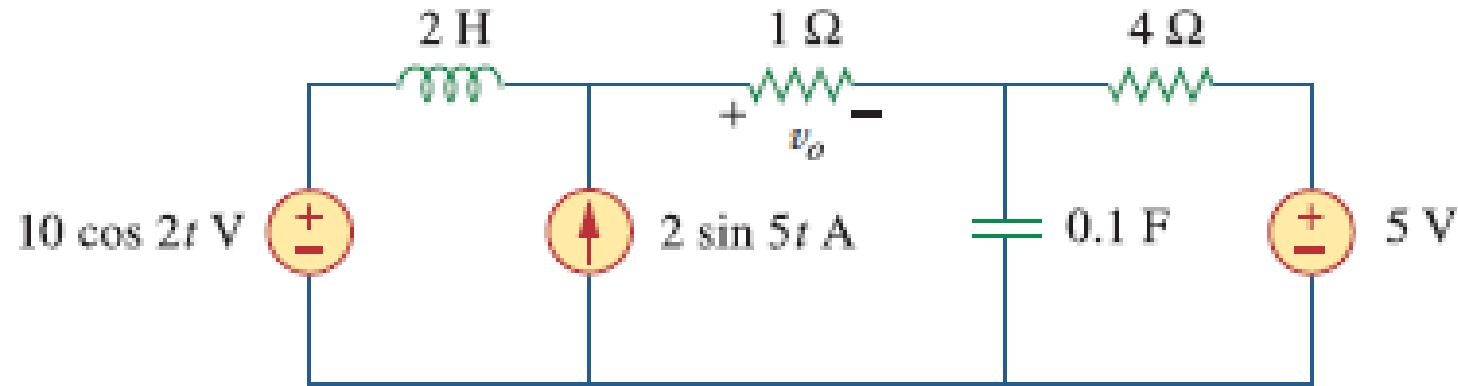
Superposition Theorem

- Superposition theorem applies to ac circuits as they are linear.
- When a circuit has sources operating at different frequencies, one must add the responses due to the individual frequencies in the time domain. (Not in the frequency domain).

The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

Numerical -1

Find v_o of the circuit using the superposition theorem.



Solution:

The circuit is operating at three different frequencies.

$$\omega = 0$$

$$\omega = 2 \text{ rad/s}$$

$$\omega = 5 \text{ rad/s}$$

$$v_o = v_1 + v_2 + v_3$$

v_1 ----- due to 5V dc source

v_2 ----- due to $10 \cos 2t$ V voltage source

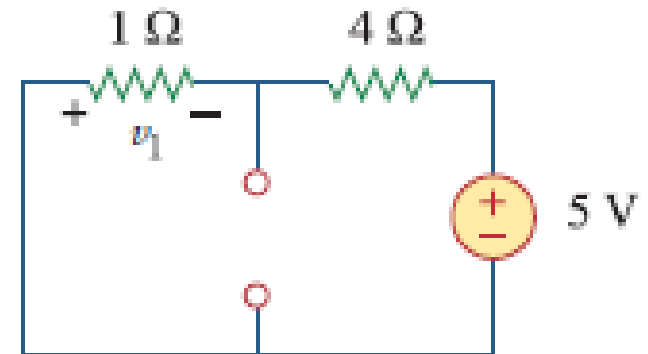
v_3 ----- due to $2 \sin 5t$ A current source

v_1 ----- due to 5V dc source:

Since $\omega = 0$, $j\omega L = 0$, $1/j\omega C = \infty$.

$$-v_1 = \frac{1}{1 + 4}(5) = 1 \text{ V}$$

$$v_1 = -1 \text{ V}$$



v_2 ----- due to $10 \cos 2t$ V voltage source:

$$10 \cos 2t \Rightarrow 10 \angle 0^\circ, \quad \omega = 2 \text{ rad/s}$$

$$2 \text{ H} \Rightarrow j\omega L = j4 \, \Omega$$

$$0.1 \text{ F} \Rightarrow \frac{1}{j\omega C} = -j5 \, \Omega$$

$$Z = 4 \parallel -j5$$

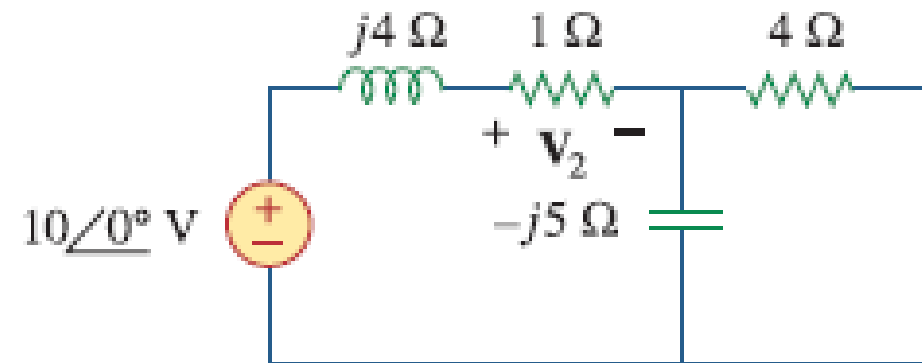
$$\mathbf{Z} = -j5 \parallel 4 = \frac{-j5 \times 4}{4 - j5} = 2.439 - j1.951$$

By voltage division,

$$\mathbf{V}_2 = \frac{1}{1 + j4 + \mathbf{Z}} (10 \angle 0^\circ) = \frac{10}{3.439 + j2.049} = 2.498 \angle -30.79^\circ$$

In the time domain,

$$v_2 = 2.498 \cos(2t - 30.79^\circ)$$

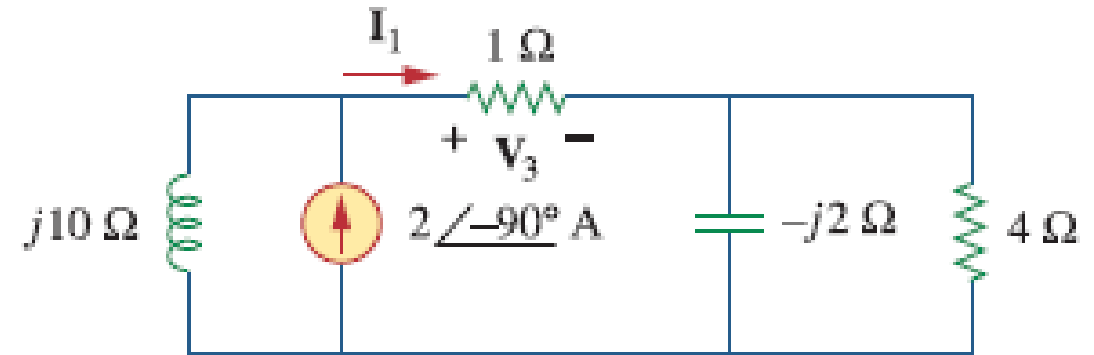


v_3 ----- due to $2 \sin 5t$ A current source:

$$2 \sin 5t \Rightarrow 2 \angle -90^\circ, \quad \omega = 5 \text{ rad/s}$$

$$2 \text{ H} \Rightarrow j\omega L = j10 \, \Omega$$

$$0.1 \text{ F} \Rightarrow \frac{1}{j\omega C} = -j2 \, \Omega$$



$$\mathbf{Z}_1 = -j2 \parallel 4 = \frac{-j2 \times 4}{4 - j2} = 0.8 - j1.6 \, \Omega$$

By current division,

$$\mathbf{I}_1 = \frac{j10}{j10 + 1 + \mathbf{Z}_1} (2 \angle -90^\circ) \text{ A}$$

$$\mathbf{V}_3 = \mathbf{I}_1 \times 1 = \frac{j10}{1.8 + j8.4} (-j2) = 2.328 \angle -80^\circ \text{ V}$$

In the time domain,

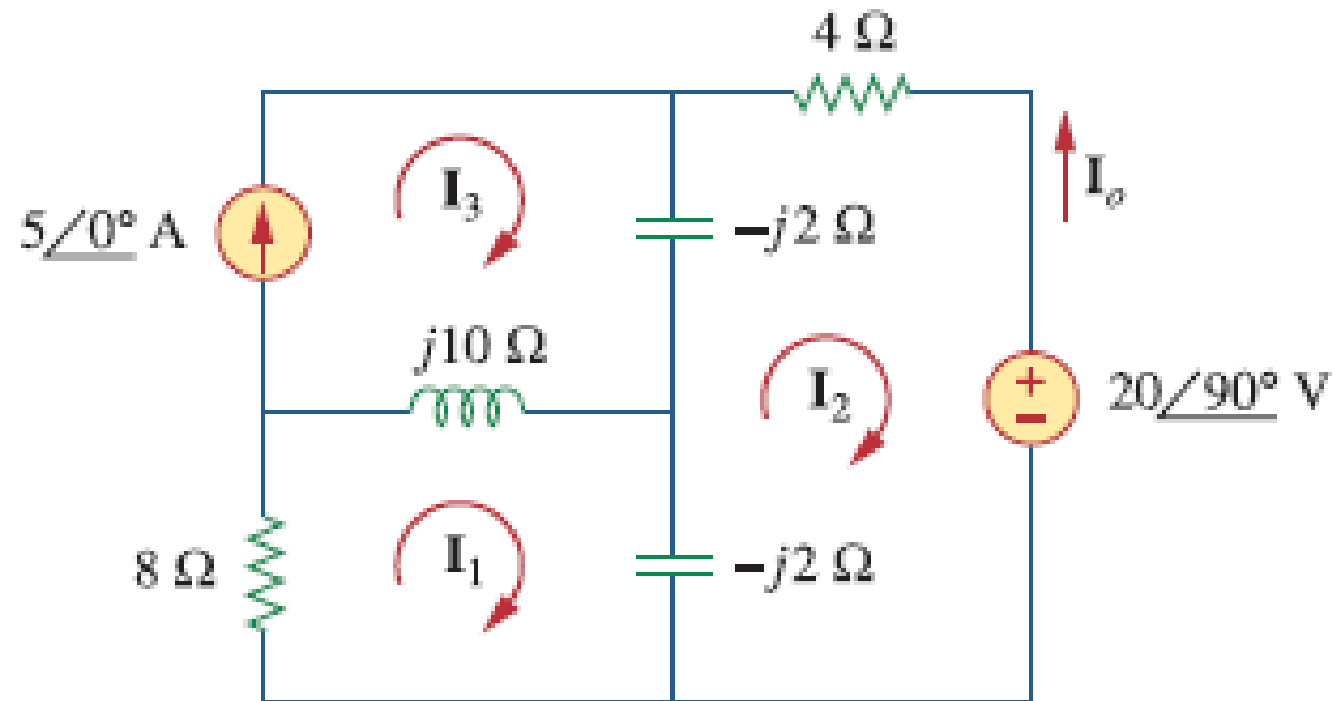
$$v_3 = 2.33 \cos(5t - 80^\circ) = 2.33 \sin(5t + 10^\circ) \text{ V}$$

$$v_o = v_1 + v_2 + v_3$$

$$v_o(t) = -1 + 2.498 \cos(2t - 30.79^\circ) + 2.33 \sin(5t + 10^\circ) \text{ V}$$

Numerical -2

Use the superposition theorem to find $\mathbf{I}_o$ in the circuit



Solution:

$$\mathbf{I}_o = \mathbf{I}_o' + \mathbf{I}_o''$$

$\mathbf{I}_o'$ ----- due to the voltage source

$\mathbf{I}_o''$ ----- due to the current source

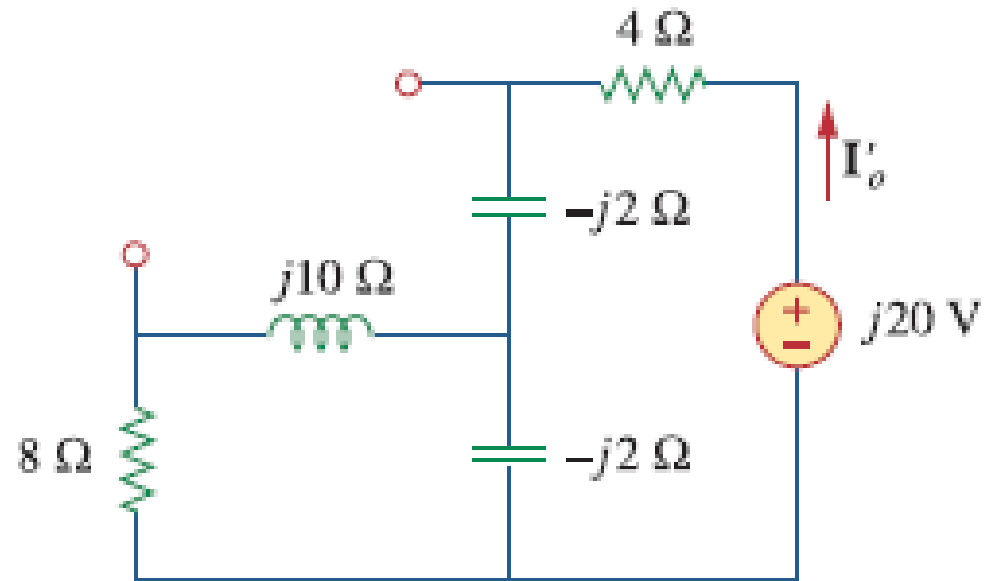
$\mathbf{I}_o'$ ----- due to the voltage source:

$$\mathbf{Z} = (8 + j10) \parallel (-j2)$$

$$\mathbf{Z} = \frac{-j2(8 + j10)}{-2j + 8 + j10} = 0.25 - j2.25$$

$$\mathbf{I}_o' = \frac{j20}{4 - j2 + \mathbf{Z}} = \frac{j20}{4.25 - j4.25}$$

$$\mathbf{I}_o' = -2.353 + j2.353$$



I_o'' ----- due to the current source:

For mesh 1,

$$(8 + j8)\mathbf{I}_1 - j10\mathbf{I}_3 + j2\mathbf{I}_2 = 0 \longrightarrow (1)$$

For mesh 2,

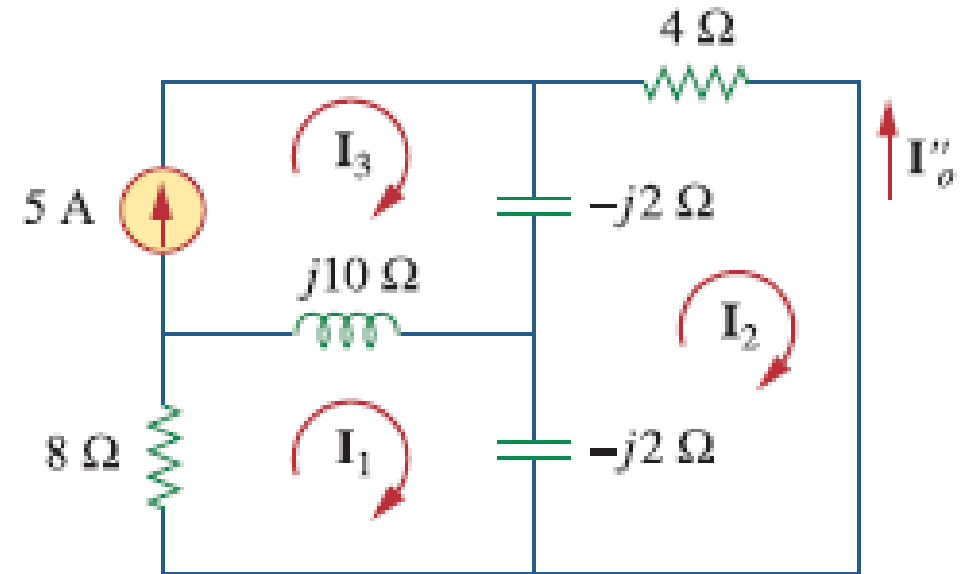
$$(4 - j4)\mathbf{I}_2 + j2\mathbf{I}_1 + j2\mathbf{I}_3 = 0 \longrightarrow (2)$$

For mesh 3,

$$\mathbf{I}_3 = 5 \longrightarrow (3)$$

Use eqn. (3) in (2),

$$(4 - j4)\mathbf{I}_2 + j2\mathbf{I}_1 + j10 = 0$$



Expressing $\mathbf{I}_1$ in terms of $\mathbf{I}_2$ gives

$$\mathbf{I}_1 = (2 + j2)\mathbf{I}_2 - 5 \longrightarrow (4)$$

Use eqn.(4) in (1),

$$(8 + j8)[(2 + j2)\mathbf{I}_2 - 5] - j50 + j2\mathbf{I}_2 = 0$$

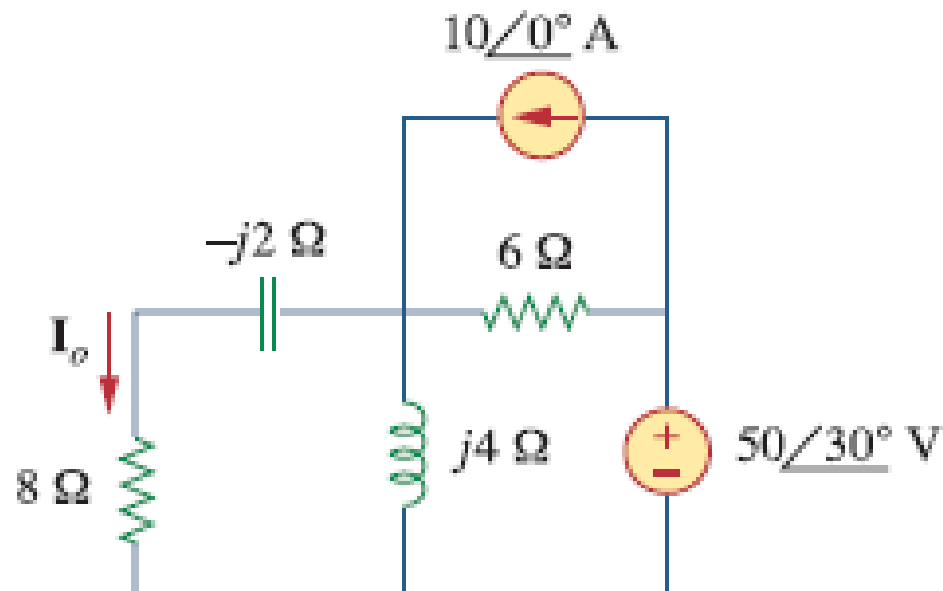
$$\mathbf{I}_2 = \frac{90 - j40}{34} = 2.647 - j1.176$$

$$\mathbf{I}_o'' = -\mathbf{I}_2 = -2.647 + j1.176$$

$$\mathbf{I}_o = \mathbf{I}_o' + \mathbf{I}_o'' = -5 + j3.529 = 6.12 \angle 144.78^\circ \text{ A}$$

Homework Problem - 1

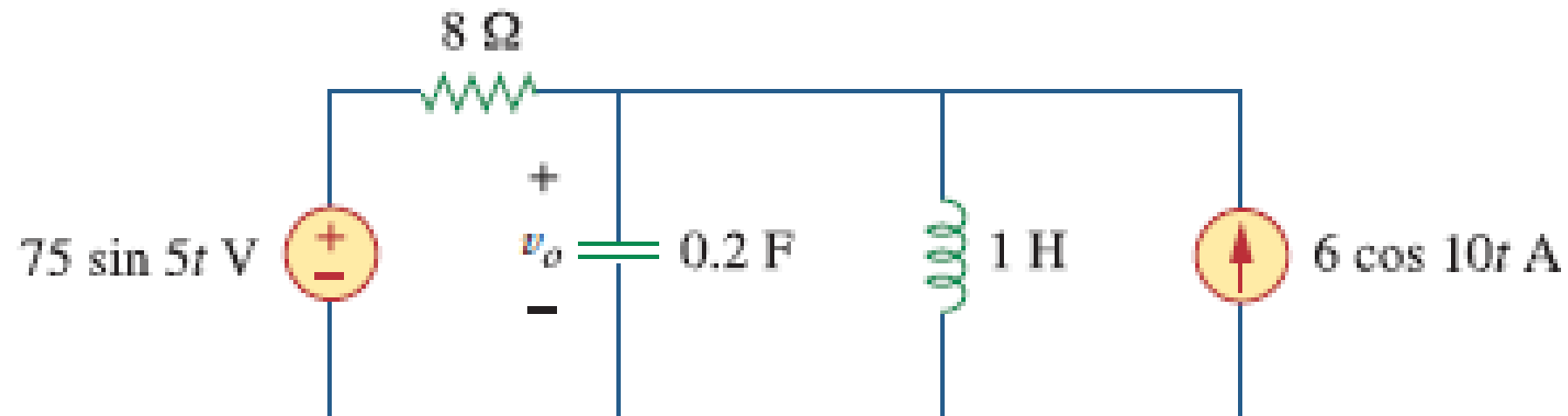
Find current $\mathbf{I_o}$ in the circuit using the superposition theorem.



Answer: $5.97\angle 65.45^\circ\text{ A}$.

Homework Problem - 2

Find v_o of the circuit using the superposition theorem.



Answer: $11.577 \sin(5t - 81.12^\circ) + 3.154 \cos(10t - 86.24^\circ) \text{ V}$.